What is a Data Warehouse?

- A physical repository where relational data are specially organized to provide enterprise-wide, cleansed data in a standardized format
- “The data warehouse is a collection of **integrated**, **subject-oriented** databases designed to support business decision making, where each unit of data is **non-volatile** and relevant to some moment in time”

Characteristics of Data Warehouses

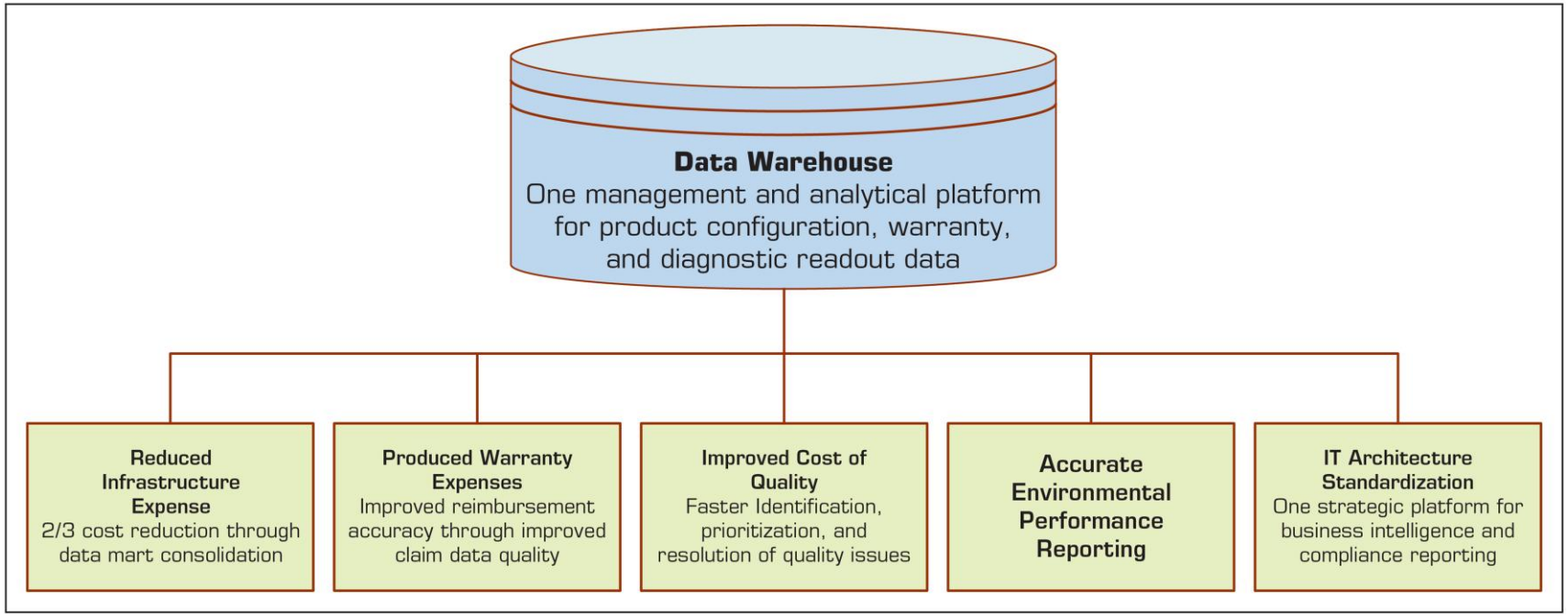
- Subject oriented
- Integrated
- Time-variant (time series)
- Nonvolatile
- Summarized
- Not normalized
- Metadata
- Relational/multi-dimensional
- Real-time/right-time/active

Data Mart

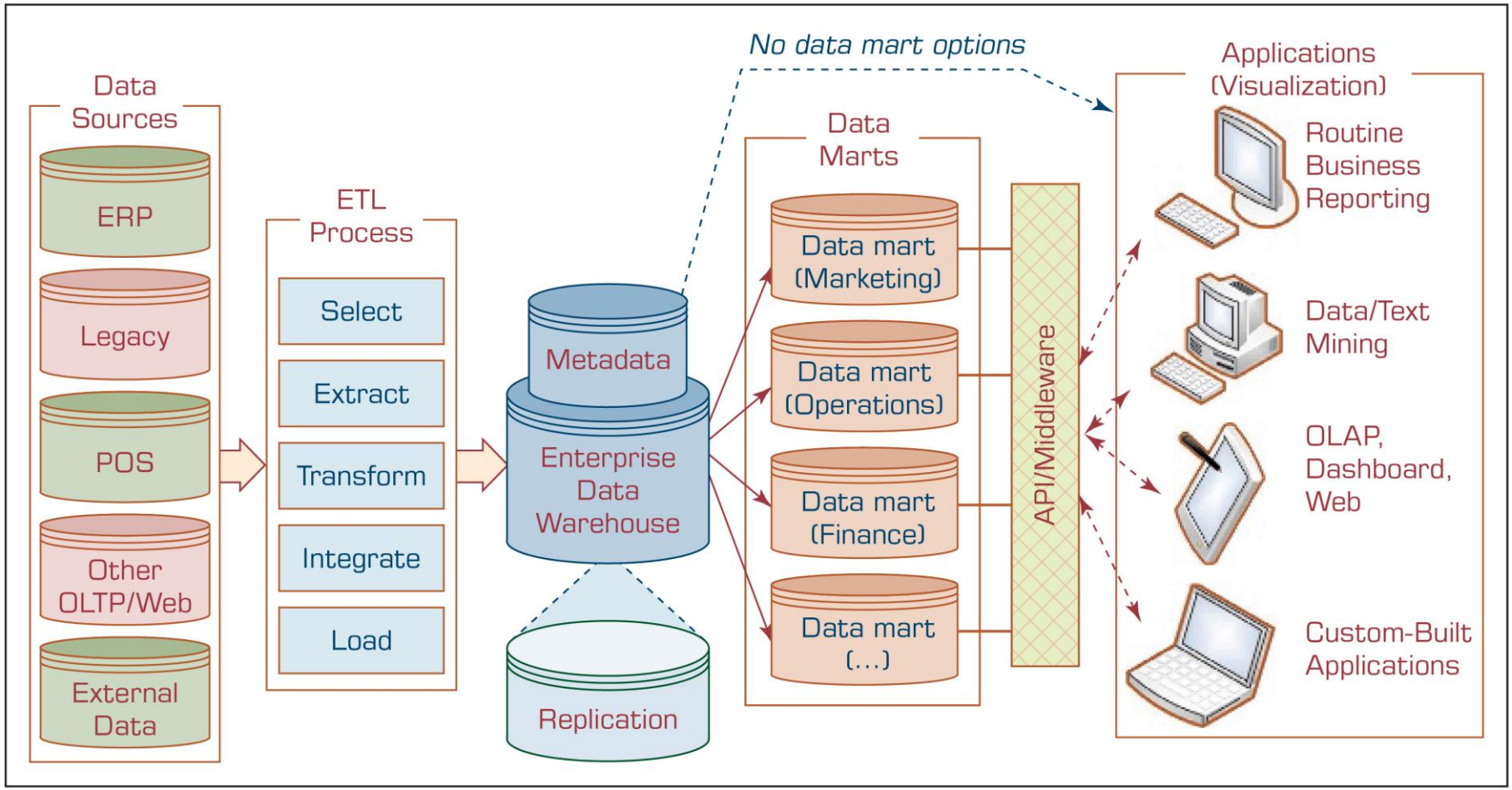
- A departmental small-scale data warehouse that stores only limited/relevant data
- **Dependent data mart**
  - A subset that is created directly from a data warehouse
- **Independent data mart**
  - A small data warehouse designed for a strategic business unit or a department

Data Warehousing for Decision Making

- An example of a data warehouse supporting data-driven decision making in automotive industry

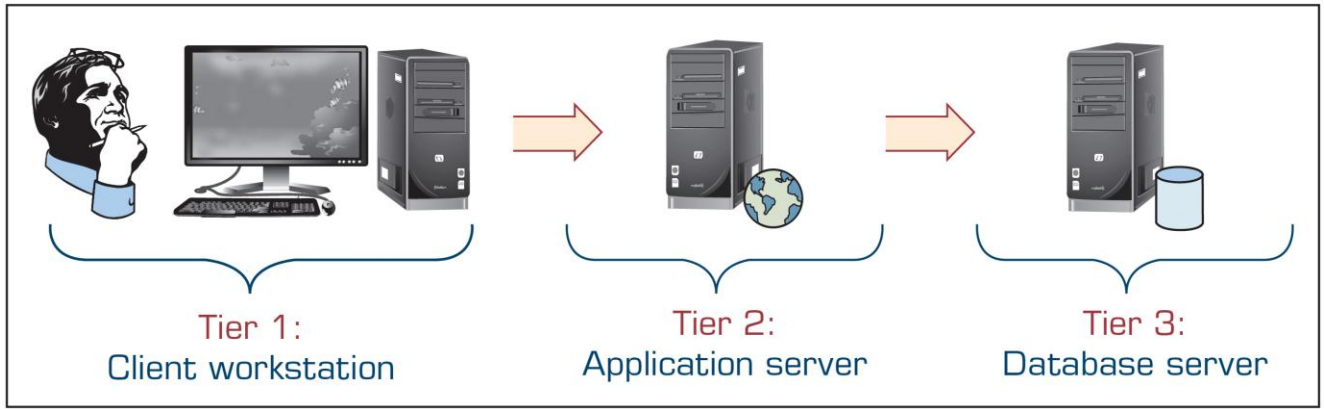


A Generic Data Warehousing Framework

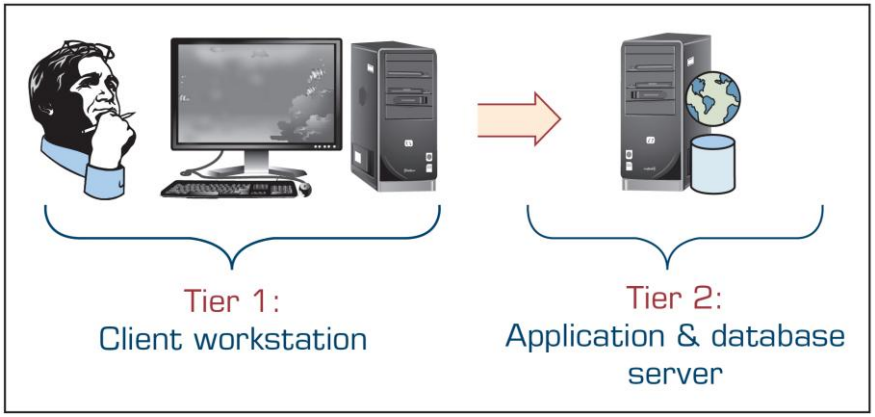


Data Warehousing Architectures

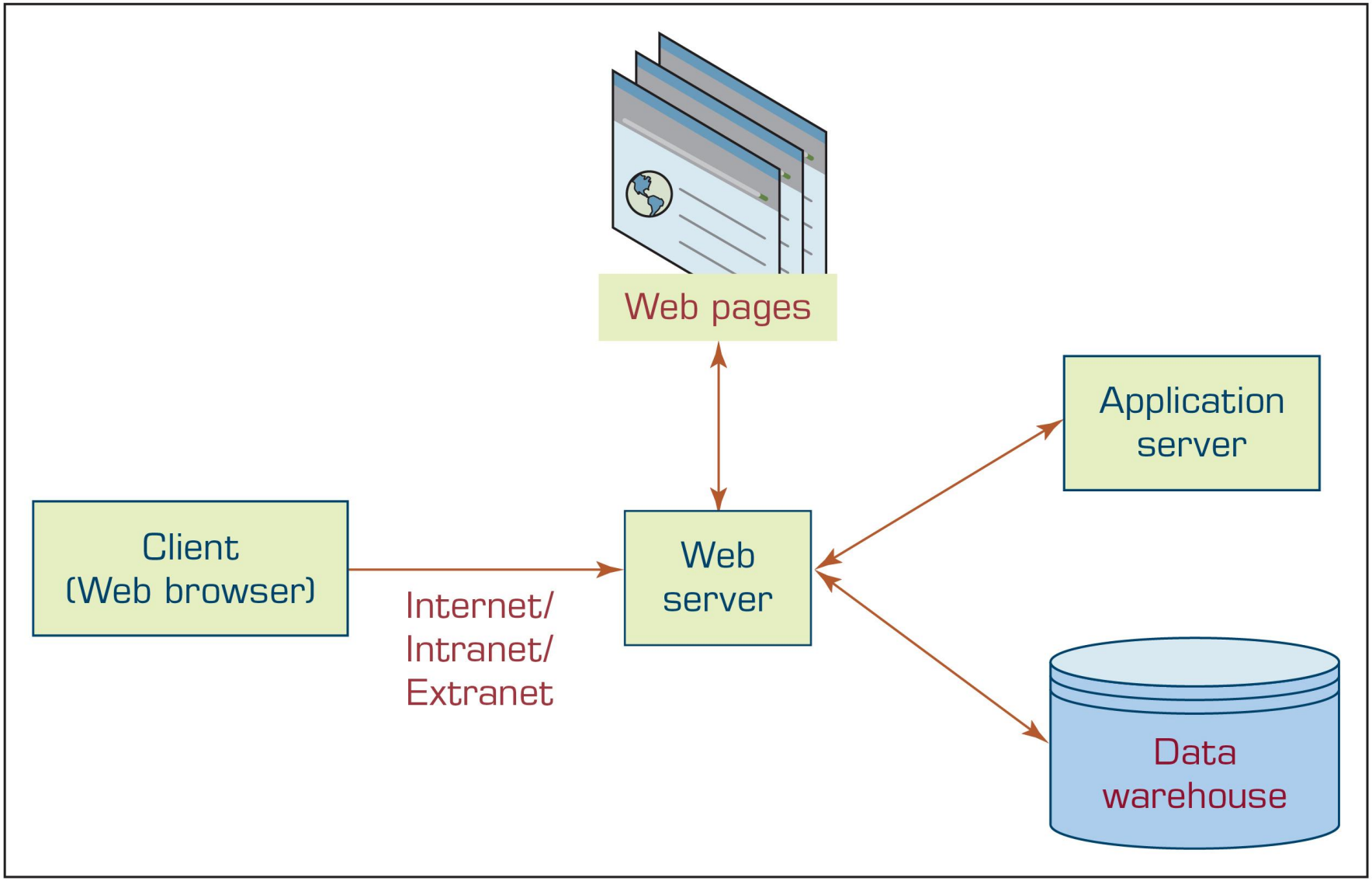
3-tier architecture



2-tier architecture



Web-based Data Warehousing Architecture



Data Modelling

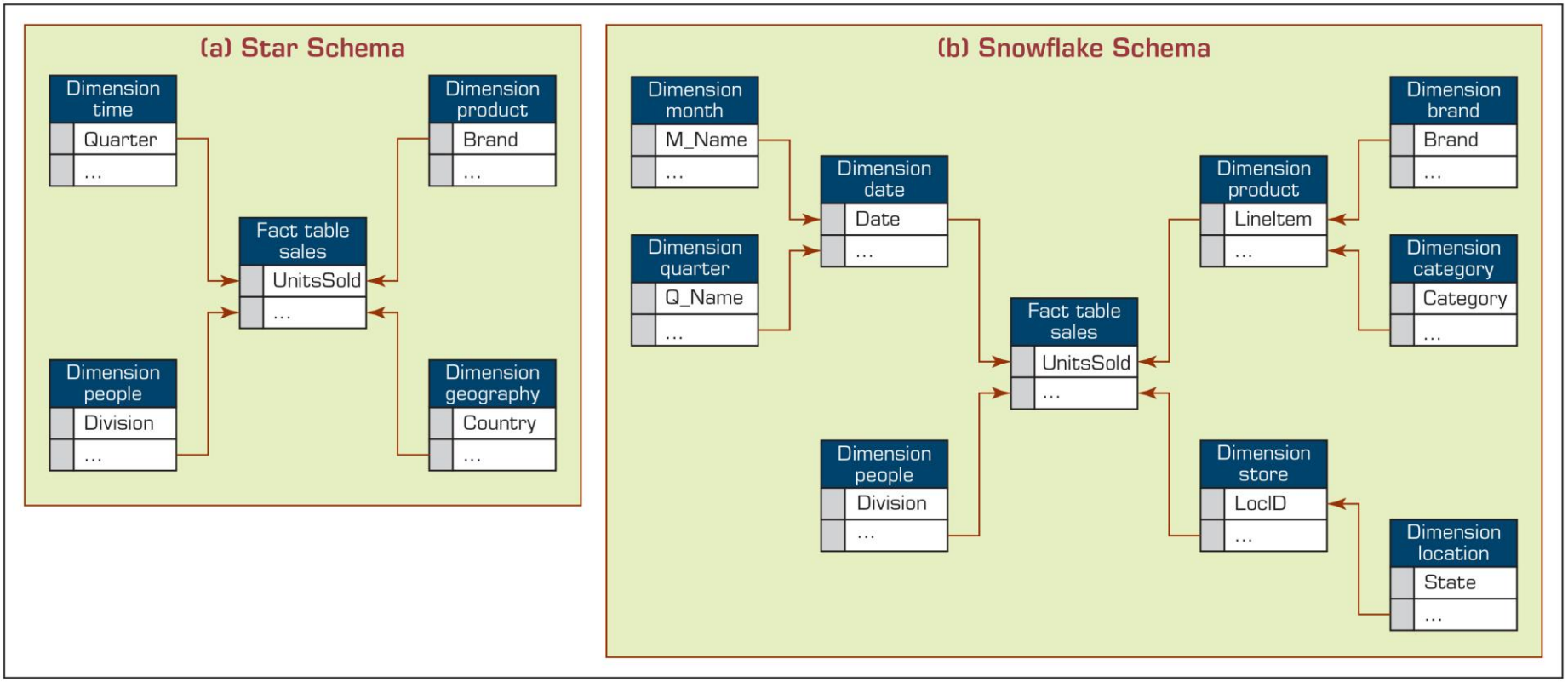
- **Dimensional Modelling**
  - A retrieval-based approach that supports high-volume query access
- **Star schema**
  - The most commonly used and the simplest style of dimensional modelling
  - Contain a **fact table** surrounded by and connected to several **dimension tables**
- **Snowflakes schema**
  - An extension of the Star schema where the diagram resembles a snowflake in shape by normalizing the dimension tables

Multidimensionality

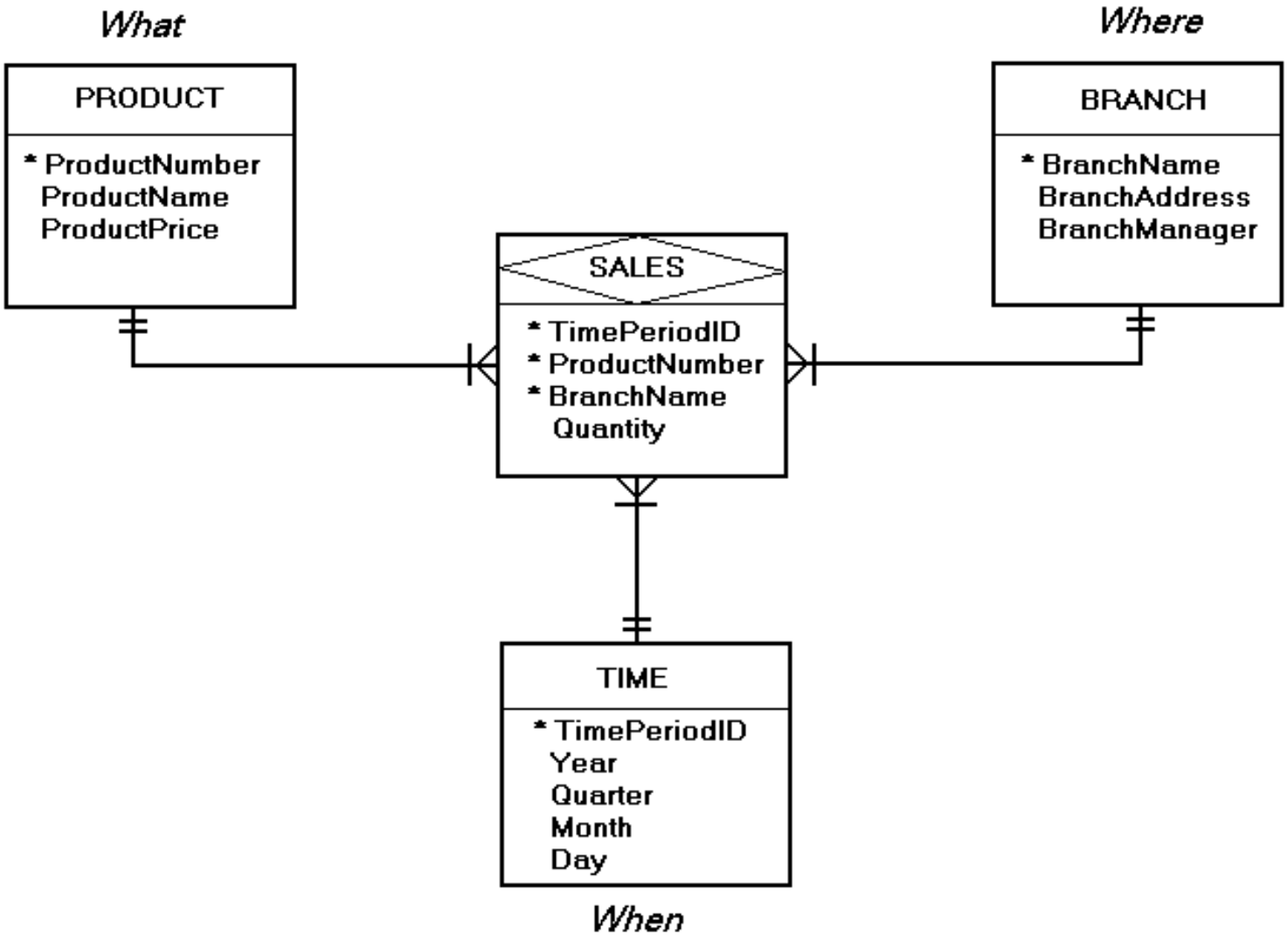
The ability to organize, present, and analyze data by several dimensions, such as sales by region, by product, by salesperson, and by time (four dimensions)

- **Multidimensional presentation**
  - **Dimensions:** products, salespeople, market segments, business units, geographical locations, distribution channels, country, or industry
  - **Measures:** money, sales volume, head count, inventory profit, actual versus forecast
  - **Time:** daily, weekly, monthly, quarterly, or yearly

Star Schema vs. Snowflake Schema

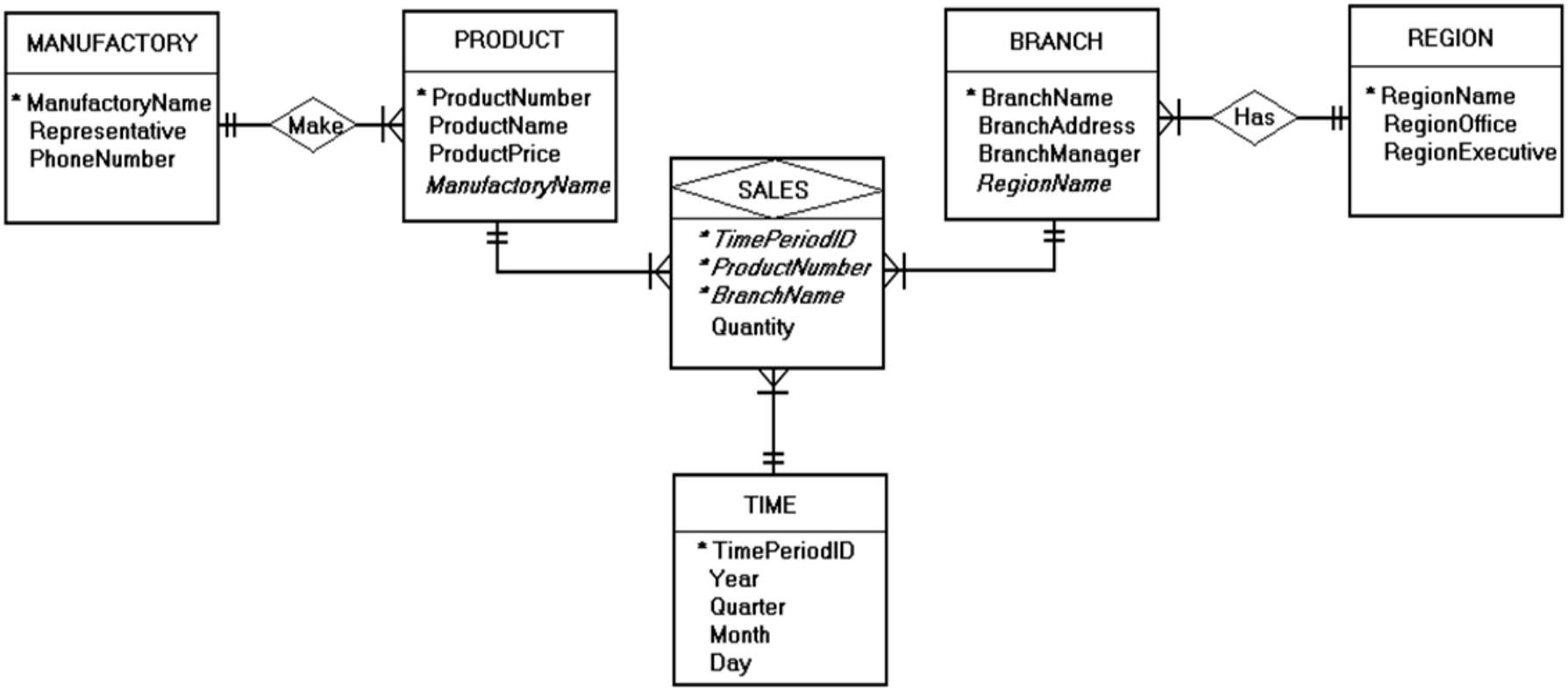


Star Schema


What
Where
PRODUCT
* ProductNumber
ProductName
ProductPrice
BRANCH
* BranchName
BranchAddress
BranchManager
SALES
* TimePeriodID
* ProductNumber
* BranchName
Quantity
TIME
* TimePeriodID
Year
Quarter
Month
Day
When

Snowflake Schema

Snowflake design has 1:M relationship chains towards the **fact table**



Data Processing

- OLTP vs. OLAP
- **OLTP** (Online **Transaction** Processing)
  - Capturing and storing data from ERP, CRM, POS
  - The main focus is on the efficiency of business transactions
- **OLAP** (Online **Analytical** Processing)
  - Converting data into information for decision support
  - Data cubes, drill-down/up, rollup, slice & dice
  - Requesting ad hoc reports
  - Conducting statistical and other analyses
- The word "**online**" means data availability

OLAP vs. OLTP

Criteria	OLTP	OLAP
Purpose	To carry out day-to-day business functions	To support decision making and provide answers to business and management queries
Data source	Transaction database (a normalized data repository primarily focused on efficiency and consistency)	Data warehouse or DM (a nonnormalized data repository primarily focused on accuracy and completeness)
Reporting	Routine, periodic, narrowly focused Reports	Ad hoc, multidimensional, broadly focused reports and queries
Resource requirements	Ordinary relational databases	Multiprocessor, large-capacity, specialized databases
Execution speed	Fast (recording of business transactions and routine reports)	Slow (resource intensive, complex, large-scale queries)

OLAP Operations

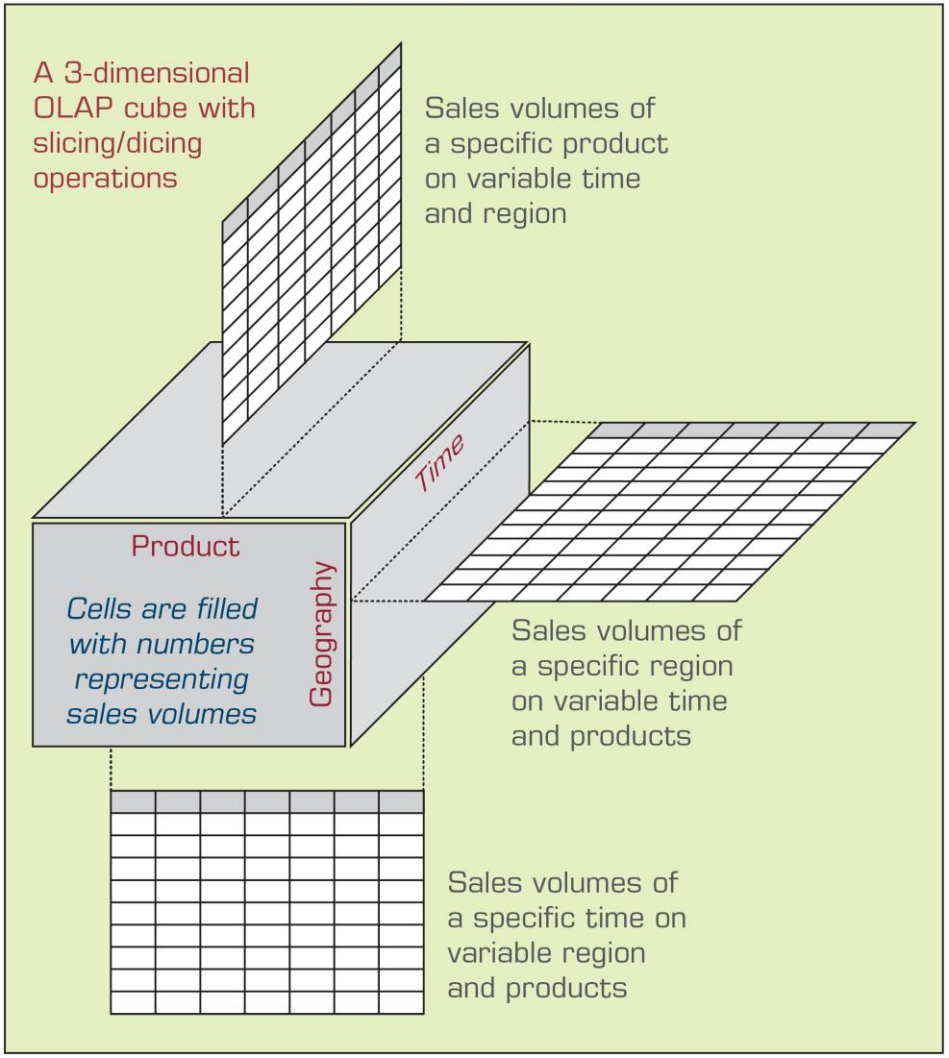
- **Data cube -** a multi-dimensional array, similar to a 3D or higher-dimensional spreadsheet, used to organize and analyze large datasets from different perspectives
- **Slice** - a subset of the data cube on one dimension
- **Dice** - a sub data cube
- **Drill Down/Up** - navigating among levels of data ranging from the most summarized (up) to the most detailed (down)
- **Rollup** - computing aggregates based on a specified hierarchy of dimensions
- **Cube** - computing aggregates for all possible combinations of the specified dimensions
- **Pivot** - computing aggregates in a cross-tabulation format (i.e., rotating the Cube results from the “long” format to the “wide” format)

Slicing

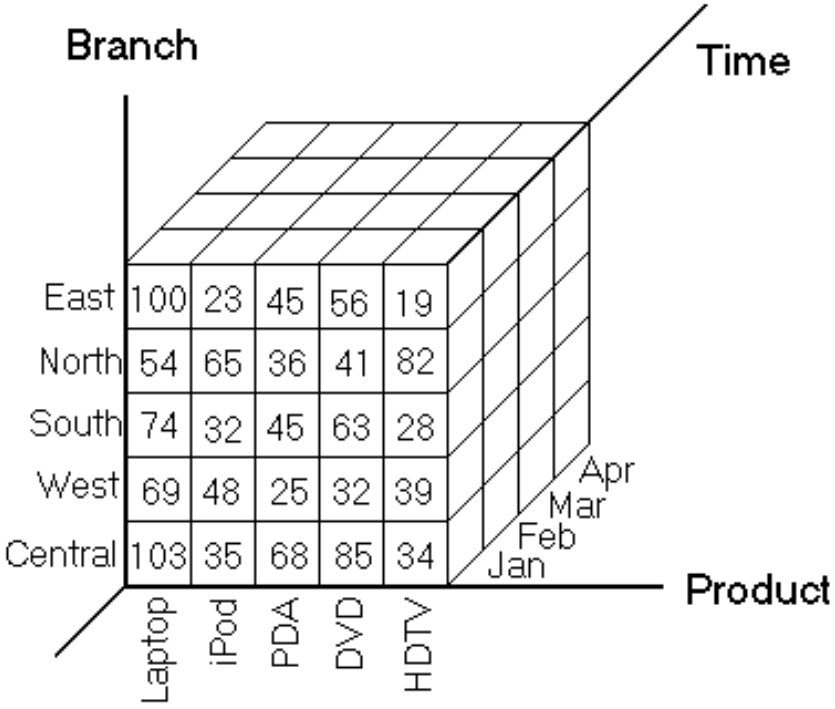
Slicing operation on a 3D data cube

- Example – an ‘iPhone17’ slice

```
SELECT Region, Time,
   SUM(SalesAmount) AS TotalSales
FROM SalesData
WHERE Product = ‘iPhone17'
GROUP BY Region, Time;
```

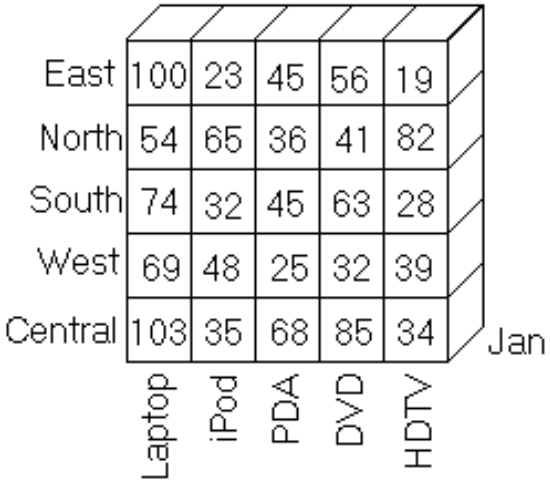


Slicing



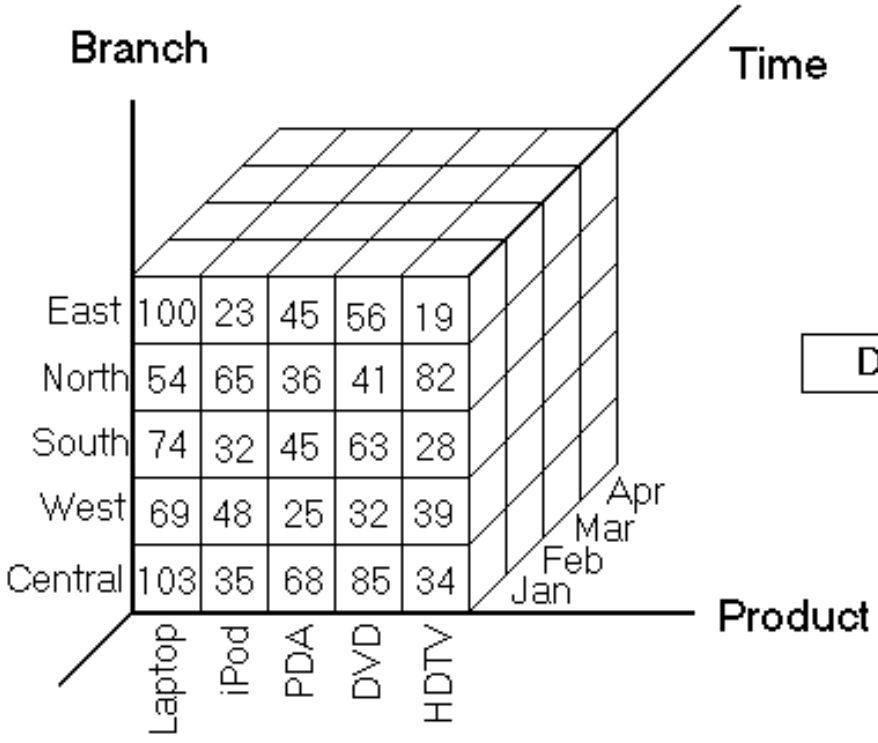
Sales data cube

Slicing

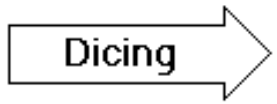


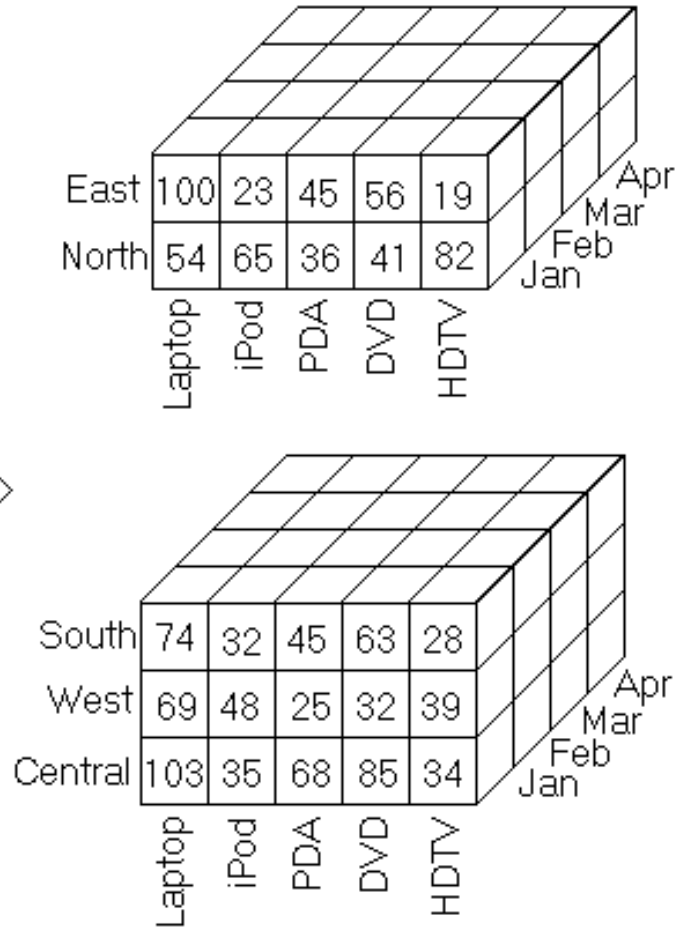
Slice of Jan.

Dicing



Sales data cube





Two Dices

Cube

```
SELECT Region, Product, SUM(Amount) AS TotalAmount
FROM Sales
GROUP BY CUBE(Region, Product);
```

Region	Product	TotalAmount
NS	iPad	1000
NS	iPhone	2000
NS	NULL	3000
NB	iPad	1000
NB	iPhone	2000
NB	NULL	3000
NULL	iPad	2000
NULL	iPhone	4000
NULL	NULL	6000

This SQL query could produce results showing:

- TotalAmount for each Region and Product combination
- Subtotals for each Region (with Product as NULL) and each Product (with Region as NULL)
- The total for all TotalAmount (with both Region and Product as NULL)

Rollup

```
SELECT Province, City, SUM(Amount) AS TotalAmount
FROM Sales
GROUP BY ROLLUP(Province, City);
```

Province	City	TotalAmount
NS	Halifax	1000
NS	Dartmouth	2000
NS	NULL	3000
NB	Moncton	1500
NB	Saint John	2500
NB	NULL	4000
NULL	NULL	7000

This SQL query could produce results showing:

- TotalAmount for each Province and City combination
- Subtotals for each Province (with City as NULL) because ROLLUP assumes there is a hierarchy Province > City
- The total for all TotalAmount (with both Province and City as NULL)

Rollup

```
SELECT Region, Product, SUM(Amount) AS TotalAmount
FROM Sales
GROUP BY ROLLUP(Region, Product);
```

Region	Product	TotalAmount
NS	iPad	1000
NS	iPhone	2000
NS	NULL	3000
NB	iPad	1000
NB	iPhone	2000
NB	NULL	3000
NULL	NULL	6000

This SQL query could produce results showing:

- TotalAmount for each Region and Product combination
- Subtotals for each Region (with Product as NULL) because ROLLUP assumes there is a hierarchy Region > Product
- The total for all TotalAmount (with both Region and Product as NULL)

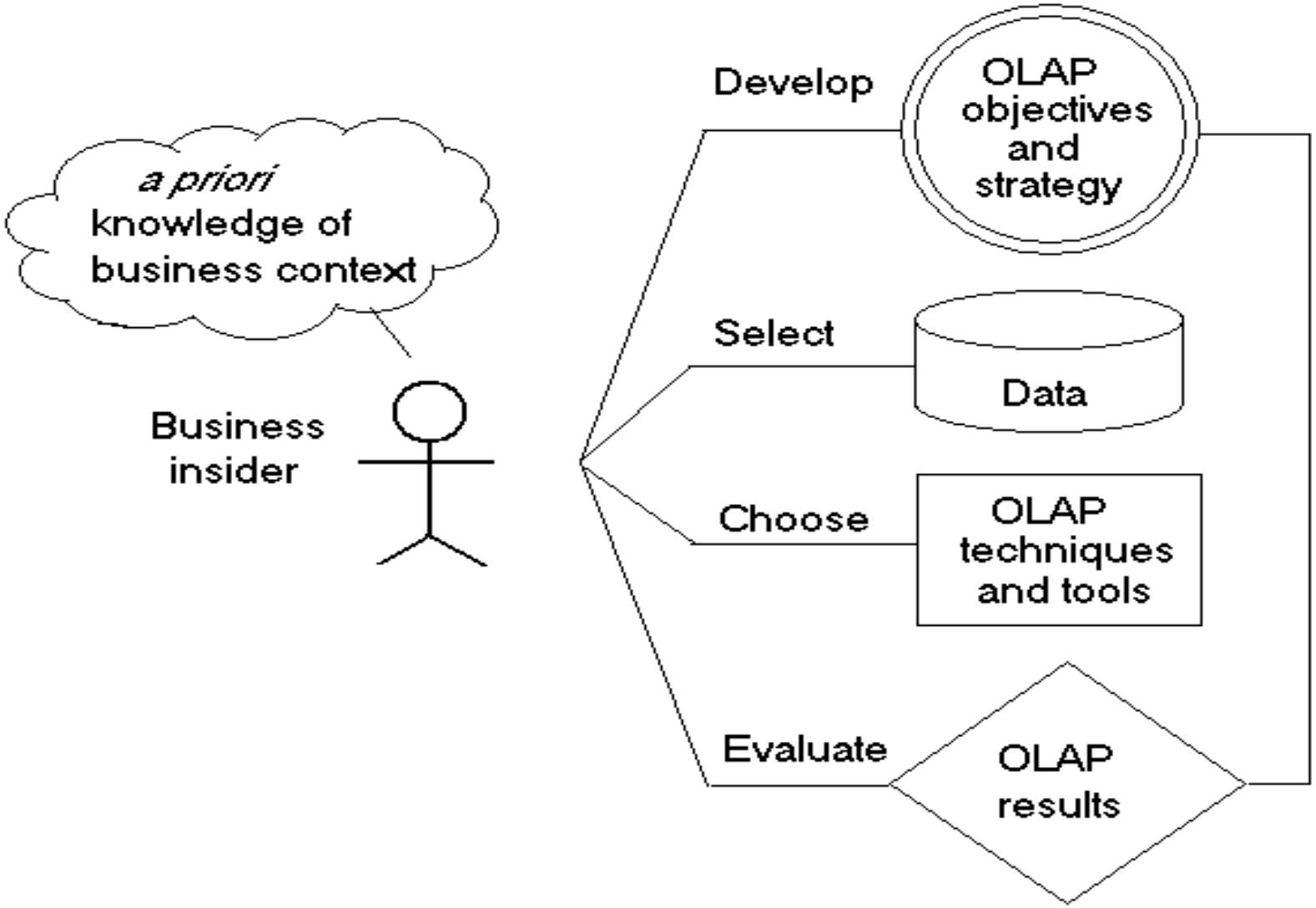
Pivot Table

		Sales				
		Product				
Branch	Month	Laptop	iPod	PDA	DVD	HDTV
East	Jan	100	23	45	56	19
	Feb	74	53	33	24	31
	Mar	28	38	26	54	29
	Apr	76	74	34	32	53
East Total		278	188	138	166	132
West	Jan	69	48	25	32	39
	Fab	49	23	43	26	38
	Mar	66	45	78	86	74
	Apr	63	13	15	47	23
West Total		247	129	161	191	174
North	Jan	54	65	36	41	82

ROLAP vs. MOLAP

- ROLAP (Relational OLAP) data warehouses
  - Storing data in relational databases, using the star/snowflake schema
  - Retrieving data using SQL
  - Offering a high level of scalability and flexibility, but with slower query times due to SQL joins
  - Examples: Oracle Data Warehouse, IBM DB2 Warehouse, Teradata, Amazon Redshift, Google BigQuery, Apache Hive
- MOLAP (Multidimensional OLAP) data warehouse
  - Storing data in pre-summarized data cubes
  - Retrieving data using specialized MDX (Multidimensional Expressions), a query language specifically designed for OLAP and data cubes
  - Offering faster query processing but with limited scalability and higher storage requirements
  - Examples: Oracle OLAP Option, Oracle Essbase, Microsoft SSAS (SQL Server Analysis Services), IBM DB2 Warehouse Cubing Services

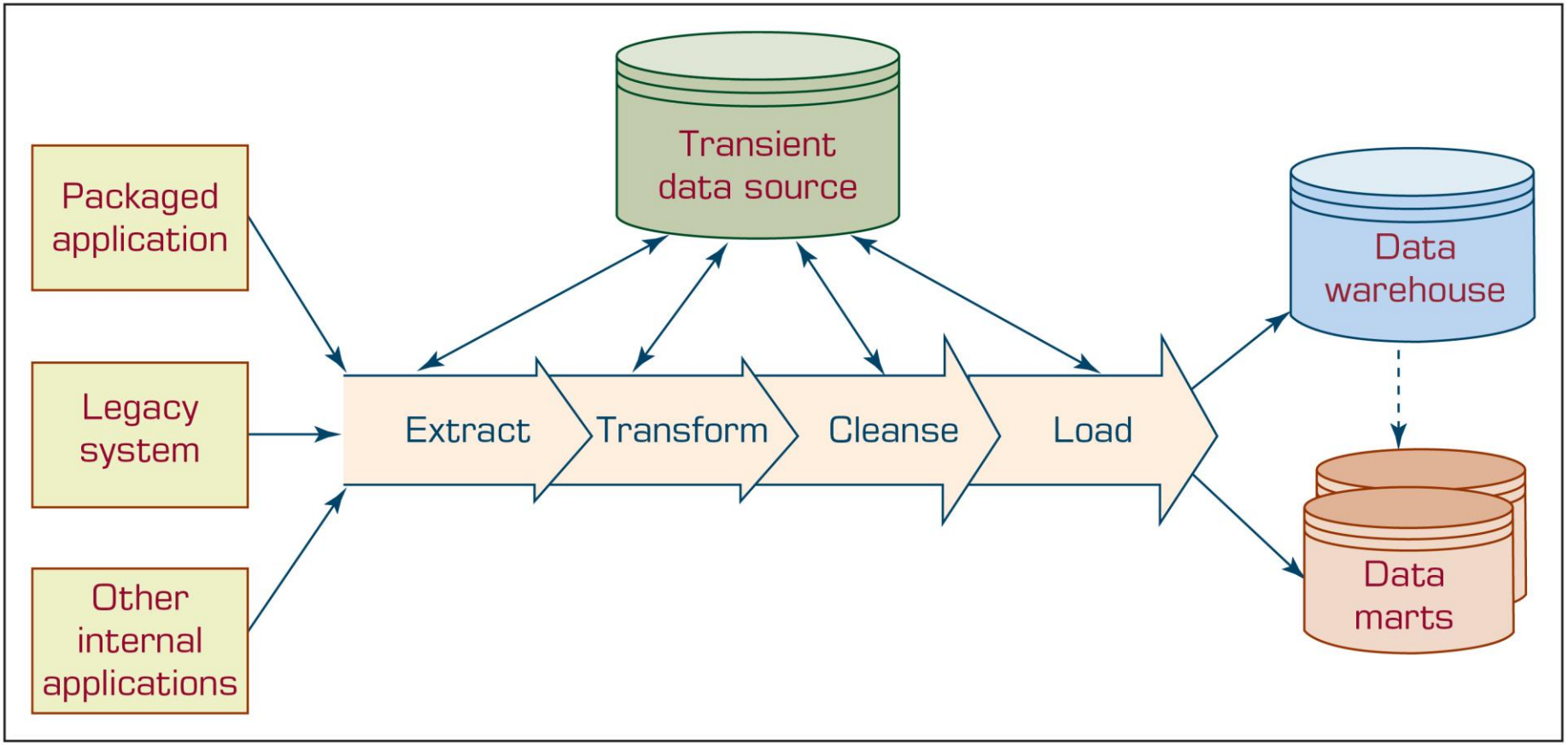
OLAP


a priori
knowledge of
business context
Business
insider
Develop
OLAP
objectives
and
strategy
Select
Data
Choose
OLAP
techniques
and tools
Evaluate
OLAP
results

Data Integration and ETL

- **ETL** = **E**xtract **T**ransform **L**oad
- **Data integration**
  - Integration that comprises three major processes: data access, data federation, and change capture.
- **Enterprise application integration (EAI)**
  - A technology that provides a vehicle for pushing data from source systems into a data warehouse
    - Focus on applications, e.g., ERP, CRM, SCM
- **Enterprise information integration (EII)**
  - An evolving tool space that promises real-time data integration from a variety of sources
    - Focus on data sources, e.g., relational databases,

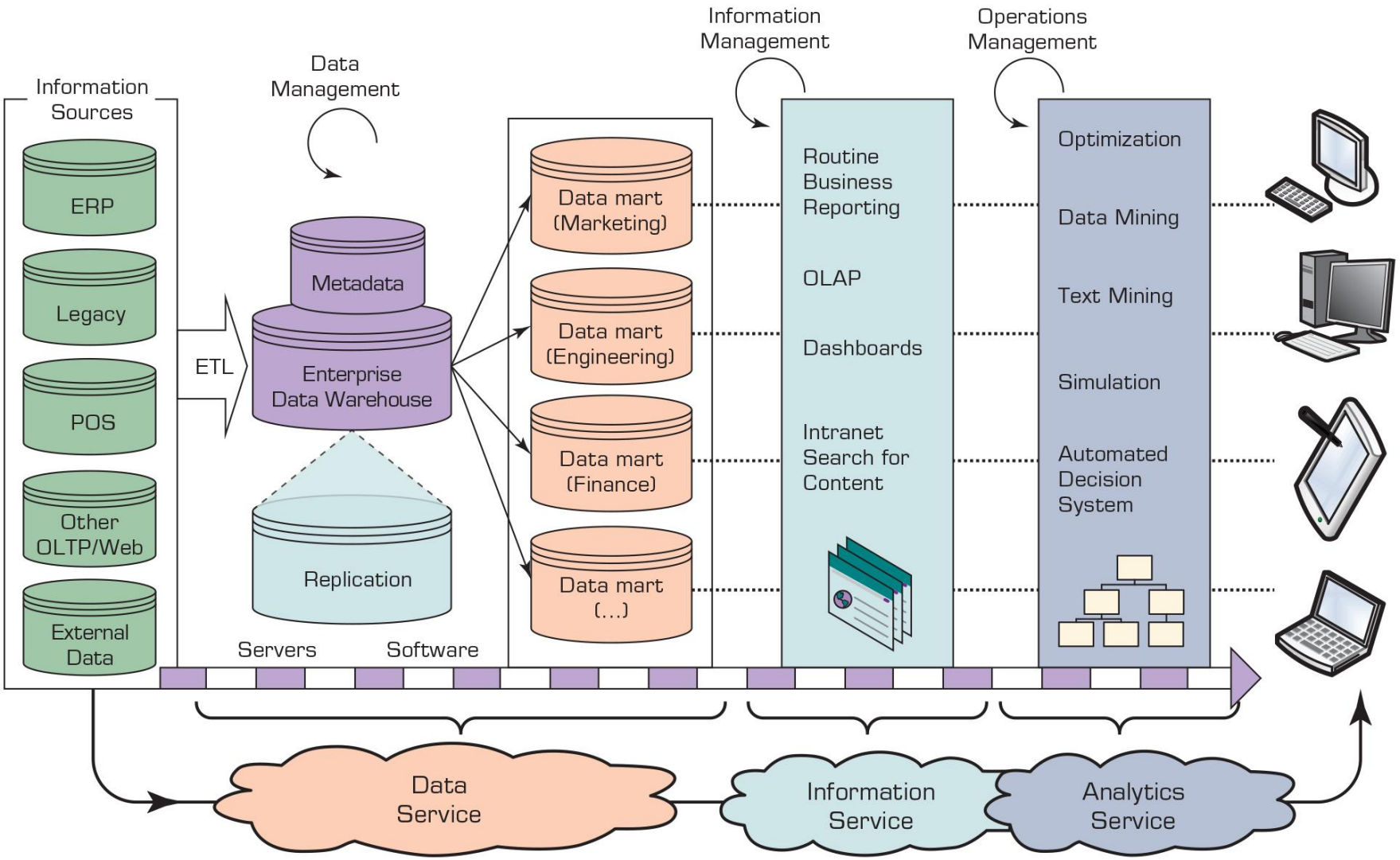
Data Integration and ETL



ETL Tools

- Examples of ETL tools
  - Microsoft SSIS (SQL Server Integration Services)
  - Oracle ODI (Oracle Data Integrator)
  - IBM InfoSphere
- Issues affecting the purchase of an ETL tool
  - Data transformation tools are expensive
  - Data transformation tools may have a long learning curve
- Important criteria in selecting an ETL tool
  - Ability to read from and write to an unlimited number of data sources/architectures
  - Automatic capturing and delivery of metadata
  - A history of conforming to open standards
  - An easy-to-use interface

Cloud-based Data Warehouse for Analytics

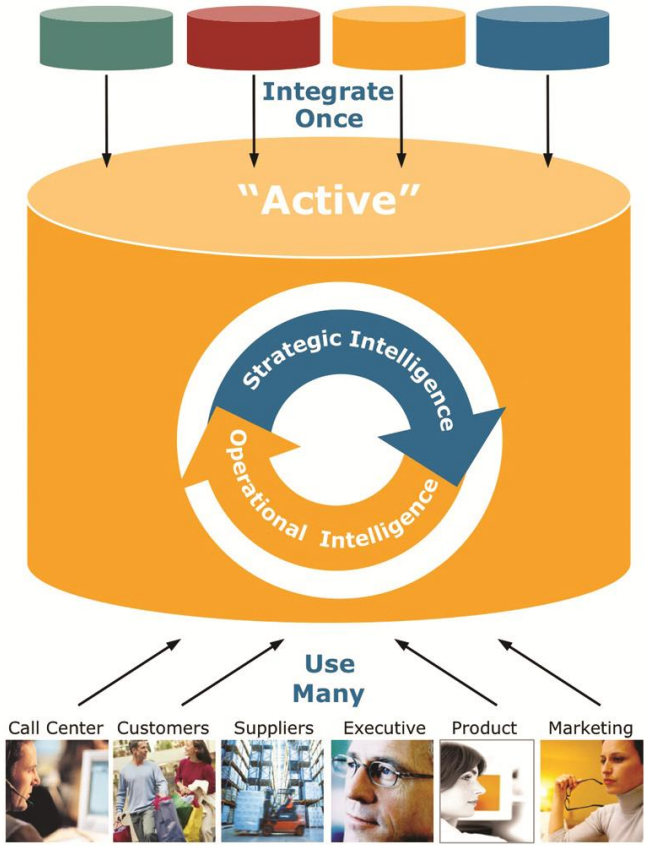


Real-Time/Active Data Warehouse

Active Access
Front-Line operational decisions or services supported by near-real-time (NRT) access; Service Level Agreements of 5 seconds or less

Active Load
Intra-day data acquisition; Mini-batch to NRT trickle data feeds measured in minutes or seconds

Active Events
Proactive monitoring of business activity initiating intelligent actions based on rules and context; to systems or users supporting an operational business process

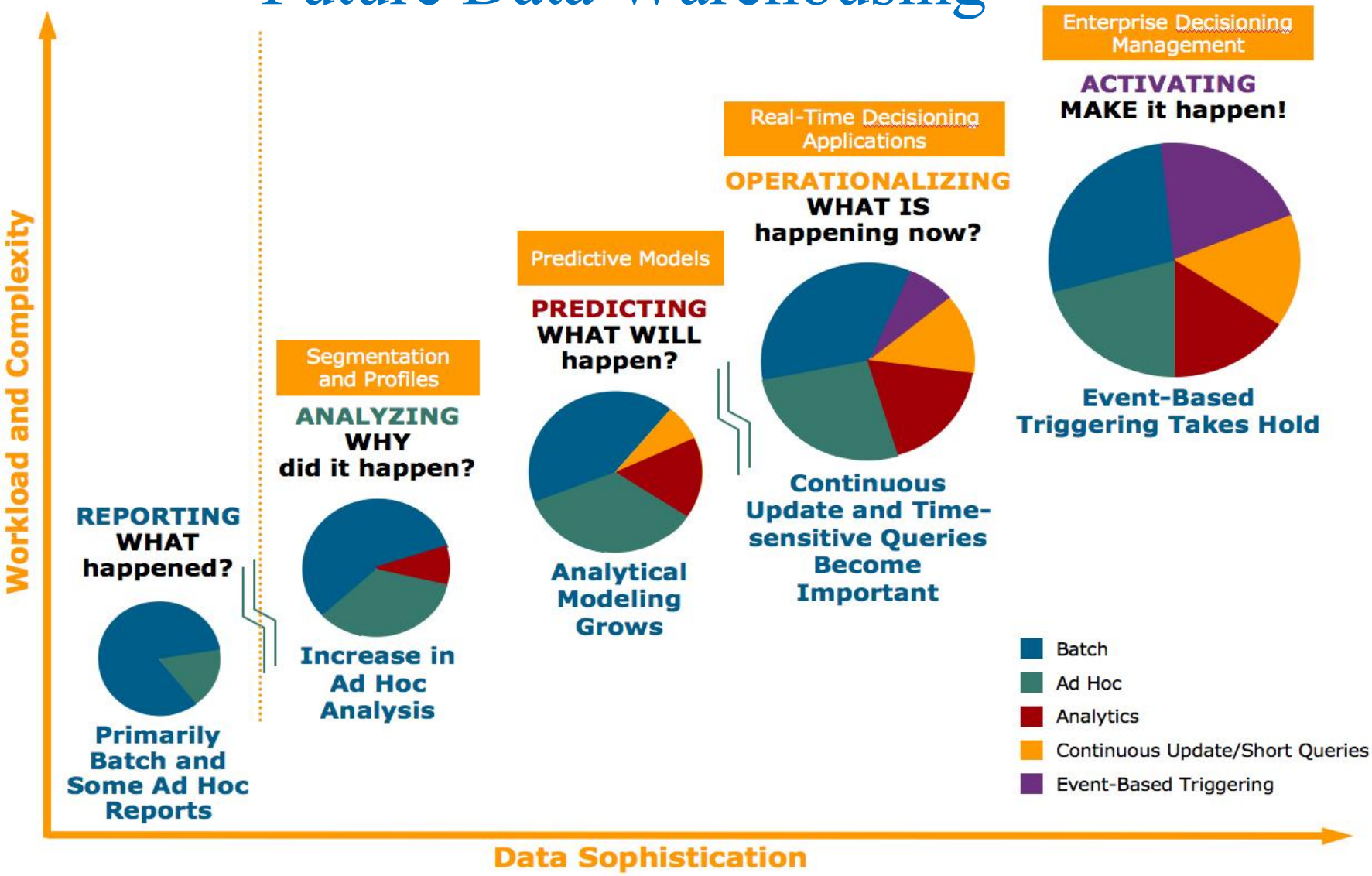


Active Workload Management
Dynamically manage system resources for optimum performance and resource utilization supporting a mixed-workload environment

Active Enterprise Integration
Integration into the Enterprise Architecture for delivery of intelligent decisioning services

Active Availability
Business Continuity to support the requirements of the business (up to 7X24)

Traditional vs. Active Data Warehouse

Traditional Data Warehouse Environment	Active Data Warehouse Environment
Strategic decisions only	Strategic and tactical decisions
Results sometimes hard to measure	Results measured with operations
Daily, weekly, monthly data currency acceptable; summaries often appropriate	Only comprehensive detailed data available within minutes is acceptable
Moderate user concurrency	High number (1,000 or more) of users accessing and querying the system simultaneously
Highly restrictive reporting used to confirm or check existing processes and patterns; often uses predeveloped summary tables or data marts	Flexible ad hoc reporting, as well as machine-assisted modeling (e.g., data mining) to discover new hypotheses and relationships
Power users, knowledge workers, internal users	Operational staffs, call centers, external users

Future Data Warehousing